Question # 10 of 10 (Start time: 01:22:48 PM, 09 January 2025) **Total Marks: 1**

Laplace transform of t^n is given by: $\mathcal{L}\{t^n\} = ?$

Select the correct option

Reload Math Equations

- $\frac{(n+1)!}{s^n}$
- $\frac{n!}{s^{n+1}}$ ✔ $\frac{n!}{s^{n+1}}$
- $\frac{(n-1)!}{s^n}$
- $\frac{n!}{s^n}$

MTH647 - Methods in Mathematical Physics (Quiz No. 3) **Quiz Start Time: 01:09 PM, 09 January 2025**

Question # 9 of 10 (Start time: 01:21:49 PM, 09 January 2025) **Total Marks:** 1

Which arithmetic operation does the scalar composition property involve?

Select the correct option

Reload Math Equations

- ○ Subtraction
- ◉ Multiplication ✔
- ○ Division
- ○ Addition

Question # 8 of 10 (Start time: 01:20:35 PM, 09 January 2025) **Total Marks: 1**

If $f(t) = \sin t$, then the Laplace transform of $f'(t)$ is:

Select the correct option

Reload Math Equations

- $\mathcal{L}(f'(t)) = \frac{-s^2}{s^2+1}$
- $\mathcal{L}(f'(t)) = \frac{-s}{s^2+1}$
- $\mathcal{L}(f'(t)) = \frac{s}{s^2+1}$ ✔ $\frac{s}{s^2+1}$
- $\mathcal{L}(f'(t)) = \frac{s^2}{s^2+1}$

Question # 7 of 10 (Start time: 01:18:54 PM, 09 January 2025) **Total Marks: 1**

Which function is defined as $\Gamma(z) = \int_0^\infty t^{z-1}e^{-t}dt$?

Select the correct option

Reload Math Equations

- Euler Function
- Laplace Function
- Gamma Function ✓
- Riemann Function

ENG 2:00 PM

Question # 5 of 10 (Start time: 01:15:45 PM, 09 January 2025) **Total Marks:** 1

In which domain is it often easier to perform algebraic manipulations for solving differential equations involving sine functions?

Select the correct option

Reload Math Equations

- ○ Fourier domain
- ○ Time domain
- ○ Frequency domain
- ◉ Laplace domain ✓

Question # 3 of 10 (Start time: 01:13:00 PM, 09 January 2025) **Total Marks:** 1

What is the Laplace transform of the second derivative $f''(t)$ of a function $f(t)$?

Select the correct option

Reload Math Equations

- $s^2F(s)-sf(0)-f'(0)$
- $s^2F(s)-sf(0)+f'(0)$
- $s^2F(s)+sf(0)+f'(0)$
- $s^2F(s)+sf(0)-f'(0)$

Question # 4 of 10 (Start time: 01:14:41 PM, 09 January 2025) **Total Marks:** 1

The condition $|f(t)e^{-kt}| \leq M$ ensures that the function is bounded for all $t \geq 0$. What does M represent in this condition?

Select the correct option

Reload Math Equations

- Minimum value of the function
- Maximum value of the function
- Lower bound on the function
- Upper bound on the function

Click to Save Answer & Move to Next Question

Question # 2 of 10 (Start time: 01:11:31 PM, 09 January 2025) **Total Marks:** 1

The expression $\frac{(n+1)!}{s^{n+2}}$ is obtained by substituting the Laplace transform of t^n in the inductive step(last step). What does this expression represent?

Select the correct option

Reload Math Equations

- ○ Laplace transform of t^{n+2}
- ○ Laplace transform of t^{2n}
- ○ Laplace transform of t^n
- ● Laplace transform of t^{n+1} ✓ t^{n+1}

Click to Save Answer & Move to Next Question

What does the term "scalar composition" refer to in the context of Laplace transforms?

Select the correct option

Reload Math Equations

- (●) Multiplying a function by a scalar constant after taking its Laplace transform. ✓
- ( ) Dividing a function by a scalar constant before taking its Laplace transform.
- ( ) Taking the derivative of a function before applying the Laplace transform.
- ( ) Taking the integral of a function before applying the Laplace transform.

Type here to search ENG 1:59 PM

Question # 1 of 10 (Start time: 04:15:46 PM, 09 January 2025)

Total Marks: 1

In the s − shifting theorem. if $F(s) = \frac{3s+2}{s^2+4s+5}$*. what is* $\mathcal{L}\{e^{-3t}f(t)\}$?

Select the correct option

Reload Math Equations

- $\frac{3(s+3)+2}{s^2+4s+5}$
- $\frac{3(s-3)+2}{(s-3)^2+4(s-3)+5}$
- $\frac{3(s-3)+2}{s^2+4s+5}$
- $\frac{3(s+3)+2}{(s+3)^2+4(s+3)+5}$ ✓

Activate Windows

Click to Save Answer & Move to Next Question

Activate Windows
Go to Settings to activate Windows.

Question # 2 of 10 (Start time: 04:18:47 PM, 09 January 2025) **Total Marks: 1**

The physical interpretation of the Laplace transform relates to:

Select the correct option

Reload Math Equations

- ○ Converting differential equations into polynomial equations.
- ◉ Mapping functions from the time domain to the frequency domain. ✔
- ○ Studying the algebraic properties of the transform.
- ○ Analyzing the geometric properties of functions.

Activate Windows

Click to Save Answer & Move to Next Question

te Windows

Question # 3 of 10 (Start time: 04:20:09 PM, 09 January 2025) **Total Marks:** 1

What is the Laplace transform of the exponential function e^{at} ?

Select the correct option

Reload Math Equations

- $L\{e^{at}\} = s/(s - a)$
- $L\{e^{at}\} = 1/(s - a)$
- $L\{e^{at}\} = 1/(s + a)$
- $L\{e^{at}\} = s/(s + a)$

Activate Windows

Click to Save Answer & Move to Next Question

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Question # 4 of 10 (Start time: 04:21:18 PM, 09 January 2025) **Total Marks:** 1

The Laplace transform of $f(t) = t^{\alpha}$, where 'α' is any constant, can be found using:

Select the correct option

Reload Math Equations

- L'Hôpital's rule
- Integration by parts
- Chain rule.
- Partial fraction decomposition

Activate Windows

Click to Save Answer & Move to Next Question

Activate Windows

Question # 7 of 10 (Start time: 04:35:36 PM, 09 January 2025) Total Marks: 1

What is the condition for the Laplace transform of **1** to converge?

Select the correct option

Reload Math Equations

- $s=0$
- No specific condition is required.
- $s<0$
- $s>0$

$s>0$

Activate Windows
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Activate Windows
Go to Settings to activate Windows.

Question # 5 of 10 (Start time: 04:22:35 PM, 09 January 2025) **Total Marks:** 1

What is the base case(1st step) when proving the Laplace transform of t^n using mathematical induction?Where n is whole number.

Select the correct option

Reload Math Equations

- $n = -1$
- $n=0$ — $n=0$
- $n=2$
- $n=1$

Activate Windows
Go to Settings to activate Windows.

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Question # 6 of 10 (Start time: 04:24:15 PM, 09 January 2025) **Total Marks:** 1

For the function $f(t)$ having the Laplace transform $\mathcal{L}(f(t))$,the relation between $F(s)$ and $F(s-a)$ is given by:

Select the correct option

Reload Math Equations

- ○ Periodic property
- ○ Convolution property
- ◉ s-Shifting property ✔
- ○ Symmetric property

Activate Windows
Activate Windows

Question # 8 of 10 (Start time: 04:37:34 PM, 09 January 2025) Total Marks: 1

What does the $s - shifting\ theorem$ state?

Select the correct option

Reload Math Equations

- The Laplace transform of $e^{-at}f(t)$ is $F(s) - a$.
- The Laplace transform of $e^{-at}f(t)$ is $F(s) \cdot a \cdot$
- The Laplace transform of $e^{-at}f(t)$ is $F(s + a)$.
- The Laplace transform $of\ e^{-at}f(t)$ is $F(s - a)$.

Activate Windows

Click to Save Answer & Move to Next Question

Windows

Question # 9 of 10 (Start time: 04:39:32 PM, 09 January 2025) Total Marks: 1

The Laplace transform converts a function of time, f(t), into a function of a complex variable, s. What does the complex variable s represent?

Select the correct option

Reload Math Equations

- (●) Frequency and growth/decay. ✓
- ( ) Energy and velocity.
- ( ) Phase and damping.
- ( ) Time and amplitude.

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Go to Settings to activate Windows.

Question # 10 of 10 (Start time: 04:40:15 PM, 09 January 2025) **Total Marks:** 1

The Laplace transform is a special case of an integral transform because:

Select the correct option

Reload Math Equations

- It maps functions from the time domain to the frequency domain.
- It uses a particular type of exponential function as its kernel. ✔
- It only works for functions defined on a finite interval.
- It is limited to solving linear differential equations.

Activate Windows
Go to Settings to activate Windows.
Activate Windows

Question # 10 of 10 (Start time: 12:57:56 PM, 10 January 2026) Total Marks: 1

What is the Laplace transform of the constant function $f(t)=7$?

Select the correct option

Reload Math Equations

- $s+7$
- $1/7s$
- $7/s$
- $7s$

k to Save Answer & Move to Next Question

$f(t) = 7$

$F(s) = 7/s$

constant function

What condition must a function ***f(t)*** satisfy for the Laplace transform to exist according to the existence theorem?

Select the correct option

Reload Math Equations

- [x] It must be piecewise continuous and satisfy a boundedness condition.
- [ ] It must be periodic with a bounded period.
- [ ] It must be continuous everywhere.
- [ ] It must be piecewise differentiable.

Click to Save Answer & Move to Next Question

The Laplace transform of the exponential function e^{at} exists for values of s when:

Select the correct option

Reload Math Equations

- ○ $s=a$
- ○ $s>a$ ✓
- ○ $s\neq a$
- ○ $s<a$

Click to Save Answer & Move to Next Question

How does the linearity property simplify the analysis of functions in Laplace transforms?

Select the correct option

Reload Math Equations

- ○ It allows for differentiation of functions.
- ○ It preserves linear combinations of functions. ✓
- ○ It allows for integration of functions.
- ○ It allows for nonlinear transformations of functions.

Click to Save Answer & Move to Next Question

Which function is defined as $\Gamma(z) = \int_0^\infty t^{z-1}e^{-t}dt$?

Select the correct option

Reload Math Equations

○	Laplace Function
○	Gamma Function ✔
○	Riemann Function
○	Euler Function

Click to Save Answer & Move to Next Question

The inverse Laplace transform of the function $1/s$ is:

Select the correct option

Reload Math Equations

○	1 ✓ 1
○	s
○	1/s
○	0

Click to Save Answer & Move to Next Question

Question # 4 of 10 (Start time: 12:55:11 PM, 10 January 2025) Total Marks: 1

Within each interval, a piecewise continuous function must be:

Select the correct option

Reload Math Equations

- ○ Differentiable
- ○ Discontinuous
- ○ Continuous ✓
- ○ Monotonic

Click to Save Answer & Move to Next Question

What is a piecewise continuous function?

Select the correct option

Reload Math Equations

- ○ A function that has a continuous derivative.
- ○ A function that has no discontinuities.
- ○ A function that is continuous everywhere. ✔
- ○ A function that is defined by different formulas on different intervals.

Click to Save Answer & Move to Next Question

What is the Laplace transform of $e^{-2t}sint$ using the $s-shifting\ theorem$,

if the Laplace transform of $sin\ (t)$ is $\frac{1}{s^2+1}$?

Select the correct option

Reload Math Equations

- ○ $\frac{1}{s^2+1}$
- ○ $\frac{1}{(s-2)^2+1}$
- ○ $\frac{1}{s^2+4}$
- ○ $\frac{1}{(s+2)^2+1}$ ✓

Click to Save Answer & Move to Next Question

What is the Laplace transform of $e^{-2t}sint$ using the $s-shifting\ theorem$,

if the Laplace transform of $sin\,(t)$ is $\frac{1}{s^2+1}$?

Select the correct option

Reload Math Equations

- $\frac{1}{(s+2)^2+1}$
- $\frac{1}{(s-2)^2+1}$
- $\frac{1}{s^2+4}$
- $\frac{1}{s^2+1}$

For the function $f(t)$ having the Laplace transform $\mathcal{L}(f(t))$,the relation between $F(s)$ and $F(s-a)$ is given by;

Select the correct option

Reload Math Equations

○	Symmetric property
○	Periodic property
○	Convolution property
○	s-Shifting property ✔

Click to Save Answer & Move to Next Question

What does the term "scalar composition" refer to in the context of Laplace transforms?

Select the correct option

Reload Math Equations

○	Taking the integral of a function before applying the Laplace transform.
○	Multiplying a function by a scalar constant after taking its Laplace transform. ✓
○	Dividing a function by a scalar constant before taking its Laplace transform.
○	Taking the derivative of a function before applying the Laplace transform.

Click to Save Answer & Move to Next Question

What is the relationship between the Laplace transform integral and the Gamma function integral?

Select the correct option

Reload Math Equations

○	They are equal
○	They are unrelated
○	They are inversely related
○	They are related by a scaling factor ✓

Click to Save Answer & Move to Next Question

What is assumed in the inductive hypothesis when proving the Laplace transform of t^n by using mathematical induction?

Select the correct option

Reload Math Equations

- The Laplace transform of t^{k+2}
- The Laplace transform of t^{k-1}
- The Laplace transform of t^k
- The Laplace transform of t^{k+1}

$-t^K$

Click to Save Answer & Move to Next Question

What is the base case(initial step) for the proof by mathematical induction for the Laplace transform of t^n?

Select the correct option

Reload Math Equations

- ○ $n = 1$
- ○ $n = -1$
- ○ $n = 0$ ✓
- ○ $n = 2$

Click to Save Answer & Move to Next Question

The Laplace transform is a special case of an integral transform because.

Select the correct option

Reload Math Equations

○	It maps functions from the time domain to the frequency domain.
○	It uses a particular type of exponential function as its kernel. ✓
○	It only works for functions defined on a finite interval.
○	It is limited to solving linear differential equations.

Click to Save Answer & Move to Next Question

The Laplace transform of $f(t) = t^3$ is given by:

Select the correct option

Reload Math Equations

- ○ $\frac{3s}{s^4}$
- ○ $\frac{6}{s^4}$ ✓
- ○ $\frac{4}{s^3}$
- ○ $\frac{3}{s^4}$

Click to Save Answer & Move to Next Question

$$L\{t^3\} = \frac{3!}{s^{3+1}}$$

$$= \frac{3 \times 2}{s^4}$$

$$= \frac{6}{s^4}$$

What is the Laplace transform of $e^{at} f(t)$ denoted as?

Select the correct option

Reload Math Equations

- ○ $F(s \times a)$
- ○ $F(s)$
- ○ $F(s + a)$
- ○ $F(s - a)$ ✓

Click to Save Answer & Move to Next Question

If $f(t) = \sin t$, then the Laplace transform of $f'(t)$ is:

Select the correct option

Reload Math Equations

- $\mathcal{L}(f'(t)) = \frac{s}{s^2+1}$ ✔
- $\mathcal{L}(f'(t)) = \frac{-s}{s^2+1}$
- $\mathcal{L}(f'(t)) = \frac{s^2}{s^2+1}$
- $\mathcal{L}(f'(t)) = \frac{-s^2}{s^2+1}$

Click to Save Answer & Move to Next Question

$$\frac{s}{s^2+1}$$

What is the base case(1st step) when proving the Laplace transform of t^n using mathematical induction?Where n is whole number.

Select the correct option

Reload Math Equations

○	***n=1***
○	***n = -1***
○	***n=0*** ✔
○	***n=2***

Click to Save Answer & Move to Next Question

The Laplace transform of the constant function 1 can be expressed as:

Select the correct option

Reload Math Equations

- ○ $L\{1\} = s$
- ○ $L\{1\} = 0$
- ○ $L\{1\} = 1$
- ○ $L\{1\} = 1/s$

Click to Save Answer & Move to Next Question

$$L\{1\} = \frac{1}{s}$$

The Gamma function is a generalization of the:

Select the correct option

Reload Math Equations

- Factorial function ✔
- Square root function
- Exponential function
- Sine function

Click to Save Answer & Move to Next Question

Question # 9 of 10 (start time: 02:15:34 PM, 10 January 2025) Total Marks: 1

What is the Laplace transform of the constant function $f(t)=7$?

Select the correct option

Reload Math Equations

- $s+7$
- $7s$
- $1/7s$
- $7/s$ ✓

Click to Save Answer & Move to Next Question

Question # 8 of 10 (start time: 02:14:17 PM, 10 January 2025) **Total Marks:** 1

The Laplace transform of e^{at}, where a is a constant, is a:

Select the correct option

Reload Math Equations

○	Rational function
○	Trigonometric function
○	Polynomial function
○	Exponential function ✔

Click to Save Answer & Move to Next Question

Question # 7 of 10 (Start time: 02:12:50 PM, 10 January 2023) Total marks: 1

In the s − shifting theorem, if $F(s) = \frac{3s+2}{s^2+4s+5}$, *what is* $\mathcal{L}\{e^{-3t}f(t)\}$?

Select the correct option

Reload Math Equations

○	$\frac{3(s-3)+2}{(s-3)^2+4(s-3)+5}$
○	$\frac{3(s+3)+2}{(s+3)^2+4(s+3)+5}$ ✓
○	$\frac{3(s+3)+2}{s^2+4s+5}$
○	$\frac{3(s-3)+2}{s^2+4s+5}$

Click to Save Answer & Move to Next Question

Question # 6 of 10 (start time: 02:11:50 PM, 10 January 2025) Total Marks: 1

What is the Laplace transform of t^a expressed in terms of Gamma function?

Select the correct option

Reload Math Equations

- $\frac{\Gamma(a+1)}{s^a}$
- $\frac{(a-1)!}{s^{a+1}}$
- $\frac{\Gamma(a+1)}{s^{a+1}}$ ✔
- $\frac{(a)!}{s^{a+1}}$

Click to Save Answer & Move to Next Question

The geometrical interpretation of the Laplace transform involves:

Select the correct option

Reload Math Equations

○	Analyzing the decay or growth rates of functions.
○	Visualizing points in the complex variable plane.
○	Converting differential equations into algebraic equations.
○	Mapping functions from the time domain to the frequency domain. ✓

Click to Save Answer & Move to Next Question

What is the base case(initial step) for the proof by mathematical induction for the Laplace transform of t^n?

Select the correct option

Reload Math Equations

- $n = -1$
- $n = 0$
- $n = 1$
- $n = 2$

Click to Save Answer & Move to Next Question

Question # 3 of 10 (Start time: 02:09:42 PM, 10 January 2025) **Total Marks:** 1

In which domain is it often easier to perform algebraic manipulations for solving differential equations involving sine functions?

Select the correct option

Reload Math Equations

- (●) Laplace domain
- ( ) Fourier domain
- ( ) Time domain
- ( ) Frequency domain

Click to Save Answer & Move to Next Question

What does the linearity property of the Laplace transform state?

Select the correct option

Reload Math Equations

○	The Laplace transform of a product of functions is equal to the product of their Laplace transforms.
○	The Laplace transform of any function is always linear.
○	The Laplace transform of a function depends on its linearity.
○	The Laplace transform of a sum of functions is equal to the sum of their Laplace transforms. ✓

Click to Save Answer & Move to Next Question

$$\therefore \quad L(f(t)) \text{ and } L(g(t)) \text{ exist.}$$

$$\Rightarrow \quad \alpha L(f(t)) + \beta L(g(t)) < \infty \quad \text{also exist. (Improper Integra}$$

LECTURE NO. 78

Corollary:

$$L[\alpha \{f(t)\} + \beta \{g(t)\}] = \alpha L\{f(t)\} + \beta L\{g(t)\}$$

If $\beta = 0$;

$$L[\alpha \{f(t)\}] = \alpha L\{f(t)\}$$

Here we use "Scalar Composition" property.

This property also used in calculus. For example:

- Linear Transformation: $T(\alpha v) = \alpha(Tv)$
- Derivative: $\frac{d}{dx}(\alpha \sin x) = \alpha \frac{d}{dx} \sin s$
- Integration: $\int c f(x)dx = c \int f(c)dx$

LECTURE NO. 79

What does the expression $s\mathcal{L}(f(t)) - f(0)$ represent in the context of Laplace transforms?

Select the correct option

Reload Math Equations

○	The Laplace transform of the integral of $f(t)$
○	The Laplace transform of the second derivative of $f(t)$
○	The Laplace transform of the original function $f(t)$
○	The Laplace transform of the derivative of $f(t)$ ✓

Click to Save Answer & Move to Next Question

What is the Laplace transform of $e^{-2t}sint$ using the $s-shifting\ theorem$,

if the Laplace transform of $sin\ (t)$ is $\frac{1}{s^2+1}$?

Select the correct option

Reload Math Equations

- $\frac{1}{s^2+4}$
- $\frac{1}{(s+2)^2+1}$
- $\frac{1}{s^2+1}$
- $\frac{1}{(s-2)^2+1}$

Click to Save Answer & Move to Next Question

What is the Laplace transform of $e^{-2t}sint$ using the $s-shifting\ theorem$,

if the Laplace transform of $sin\ (t)$ is $\frac{1}{s^2+1}$?

Select the correct option

Reload Math Equations

- $\frac{1}{s^2+4}$
- $\frac{1}{(s+2)^2+1}$
- $\frac{1}{s^2+1}$
- $\frac{1}{(s-2)^2+1}$

Click to Save Answer & Move to Next Question

What is the relationship between the Laplace transform integral and the Gamma function integral?

Select the correct option

Reload Math Equations

- ○ They are inversely related
- ○ They are equal
- ○ They are related by a scaling factor ✓
- ○ They are unrelated

Click to Save Answer & Move to Next Question

What is the final step in finding the Laplace transform of cosh(at)?

Select the correct option

Reload Math Equations

○	Evaluate the integral with respect to **t**
○	Replace cosh(at) with its definition.
○	Apply the limits of integration.
◉	Simplify the resulting expression. ✔

Click to Save Answer & Move to Next Question

If $f(t) = \sin t$, then the Laplace transform of $f'(t)$ is:

Select the correct option

Reload Math Equations

- $\mathcal{L}(f'(t)) = \frac{-s}{s^2+1}$
- $\mathcal{L}(f'(t)) = \frac{-s^2}{s^2+1}$
- $\mathcal{L}(f'(t)) = \frac{s}{s^2+1}$ ✔
- $\mathcal{L}(f'(t)) = \frac{s^2}{s^2+1}$

Click to Save Answer & Move to Next Question

$$\frac{s}{s^2+1}$$

The geometrical interpretation of the Laplace transform involves:

Select the correct option

Reload Math Equations

○	Visualizing points in the complex variable plane. ✔
○	Converting differential equations into algebraic equations.
○	Analyzing the decay or growth rates of functions.
○	Mapping functions from the time domain to the frequency domain.

Click to Save Answer & Move to Next Question

Question # 5 of 10 (Start time: 02:21:23 PM, 10 January 2025) **Total Marks:** 1

What happens to the amplitude of the sine function in the Laplace domain?

Select the correct option

Reload Math Equations

- ○ It remains the same
- ○ It increases
- ○ It depends on the value of s ✓
- ○ It decreases

Click to Save Answer & Move to Next Question

The Laplace transform of $f(t) = t^3$ is given by:

Select the correct option

Reload Math Equations

- ○ $\frac{3s}{s^4}$
- ○ $\frac{3}{s^4}$
- ○ $\frac{6}{s^4}$ ✓
- ○ $\frac{4}{s^3}$

Click to Save Answer & Move to Next Question

In which domain is it often easier to perform algebraic manipulations for solving differential equations involving sine functions?

Select the correct option

Reload Math Equations

- ○ Fourier domain
- ○ Time domain
- ○ Frequency domain
- ○ Laplace domain

Click to Save Answer & Move to Next Question

In which domain is it often easier to perform algebraic manipulations for solving differential equations involving sine functions?

Select the correct option

Reload Math Equations

- ○ Fourier domain
- ○ Time domain
- ○ Frequency domain
- ○ Laplace domain ✓

Click to Save Answer & Move to Next Question

Which of the following statement is true about the Laplace transform?

Select the correct option

Reload Math Equations

- (●) The Laplace transform is an integral transform.
- ( ) The Laplace transform converts complex functions into simple polynomial expressions.
- ( ) The Laplace transform is a derivative transform.
- ( ) The Laplace transform is a summation transform.

Click to Save Answer & Move to Next Question

Question # 1 of 10 (Start time: 02:18:45 PM, 10 January 2025) **Total Marks:** 1

Which of the following best describes the Laplace transform?

Select the correct option

Reload Math Equations

- ○ A method for calculating limits of functions.
- ○ A transformation that converts functions into their inverses.
- ✓ A mathematical tool for solving linear ordinary and partial differential equations.
- ○ A technique for converting complex numbers into real numbers.

Click to Save Answer & Move to Next Question

What is the base case(initial step) for the proof by mathematical induction for the Laplace transform of t^n?

Select the correct option

Reload Math Equations

- ○ $n = -1$
- ○ $n = 0$ ✔
- ○ $n = 1$
- ○ $n = 2$

Click to Save Answer & Move to Next Question

What condition must a function $f(t)$ satisfy for the Laplace transform to exist according to the existence theorem?

Select the correct option

Reload Math Equations

- (●) It must be piecewise continuous and satisfy a boundedness condition. ✓
- ( ) It must be piecewise differentiable.
- ( ) It must be periodic with a bounded period.
- ( ) It must be continuous everywhere.

Saving...

What is the Laplace transform of the second derivative $f''(t)$ of a function $f(t)$?

Select the correct option

Reload Math Equations

- $s^2F(s)-sf(0)+f(0)$
- $s^2F(s)-sf(0)-f(0)$ ✓
- $s^2F(s)+sf(0)-f(0)$
- $s^2F(s)+sf(0)+f(0)$

Click to Save Answer & Move to Next Question

$$L(f') = L(f'(t)) = \int_0 e^{-st} f'(t)\, dt$$

$f(t)$ is continuous. Integrating and applying limits, we have

$$L(f') = |e^{-st} f(t)|_0^\infty - \int_0^\infty f(t)\,(-s)e^{-st}\, dt$$

$$L(f') = \left|\lim_{t\to\infty} \frac{f(t)}{e^{st}} - \frac{f(0)}{e^0}\right| + s\int_0^\infty e^{-st} f'(t)\, dt$$

$$L(f') = 0 - f(0) + sL(f)$$

$$L(f') = sL(f) - f(0) \quad \ldots(1)$$

g derivative of equation (1),

$$L(f'') = sL(f') - f'(0)$$

ng value from equation (1),

$$L(f'') = s[sL(f) - f(0)] - f'(0)$$

$$L(f'') = s^2L(f) - sf(0) - f'(0)$$ ✓

e prove.

What does the linearity property of the Laplace transform state?

Select the correct option

Reload Math Equations

◉	The Laplace transform of a sum of functions is equal to the sum of their Laplace transforms.
○	The Laplace transform of a function depends on its linearity.
○	The Laplace transform of any function is always linear.
○	The Laplace transform of a product of functions is equal to the product of their Laplace transforms.

Saving...

Which of the following best describes the Laplace transform?

Select the correct option

Reload Math Equations

○	A technique for converting complex numbers into real numbers.
○	A method for calculating limits of functions.
○	A transformation that converts functions into their inverses.
◉	A mathematical tool for solving linear ordinary and partial differential equations. ✔

Saving...

Which of the following represents the scalar composition property of the Laplace transform?

Select the correct option

- ○ $L[af(t)]=a+L[f(t)]$
- ◉ $L[af(t)]=a\cdot L[f(t)]$ ✓
- ○ $[af(t)]=L[a]+L[f(t)]$
- ○ $L[af(t)]=L[a]\cdot L[f(t)]$

Click to Save

Required result proved.

$$\therefore \quad L(f(t)) \text{ and } L(g(t)) \text{ exist.}$$

$$\Rightarrow \quad \alpha L(f(t)) + \beta L(g(t)) < \infty \quad \text{also exist. (Improper Int}$$

LECTURE NO. 78

Corollary:

$$L[\alpha \{f(t)\} + \beta \{g(t)\}] = \alpha L\{f(t)\} + \beta L\{g(t)\}$$

If $\beta = 0$;

$$L[\alpha \{f(t)\}] = \alpha L\{f(t)\}$$

Here we use "Scalar Composition" property.

This property also used in calculus. For example:

Linear Transformation: $T(\alpha v) = \alpha(Tv)$

Derivative: $\frac{d}{dx}(\alpha \sin x) = \alpha \frac{d}{dx} \sin s$

Integration: $\int c f(x)dx = c \int f(x)dx$

The condition $|f(t)e^{-kt}| \leq M$ ensures that the function is bounded by a constant value. What is the significance of this boundedness?

Select the correct option

Reload Math Equations

- ○ It guarantees the continuity of the function.
- ◉ It ensures the existence of the Laplace transform. ✓
- ○ It guarantees the differentiability of the function.
- ○ It ensures the periodicity of the function.

Click to Save Answer & Move to Next Question

How does the Laplace transform of a constant function vary with the value of the constant ***c***?

Select the correct option

Reload Math Equations

- ○ It decreases proportionally with c.
- ◉ It increases proportionally with c. ✔
- ○ It is inversely proportional to c.
- ○ It remains constant regardless of the value of c.

Click to Save Answer & Move to Next Question

What happens to the amplitude of the sine function in the Laplace domain?

Select the correct option

Reload Math Equations

- ○ It remains the same
- ○ It increases
- ○ It depends on the value of s ✔
- ○ It decreases

Click to Save Answer & Move to Next Question

What is a piecewise continuous function?

Select the correct option

Reload Math Equations

- A function that has no discontinuities.
- A function that is defined by different formulas on different intervals. ✔
- A function that has a continuous derivative.
- A function that is continuous everywhere.

Click to Save Answer & Move to Next Question

ntegrating and applying limits, we have

$$L(1) = \left| \lim_{T\to\infty} \frac{e^{-st}}{-s} \right|_0^T \Rightarrow \left| -\frac{1}{s} \lim_{T\to\infty} e^{-st} \right|_0^T \Rightarrow -\frac{1}{s} \left| \lim_{T\to\infty} \frac{1}{e^{st}} \right|_0^T \quad \ldots(1)$$

$$L(1) = -\frac{1}{s}\left[\lim_{T\to\infty} \frac{1}{e^{sT}} - \frac{1}{e^{s(0)}}\right] \Rightarrow -\frac{1}{s}\left[\frac{1}{e^{\infty}} - \frac{1}{1}\right]$$

$$L(1) = \frac{1}{s}$$

his is our required result for $s > 0$. What if $s < 0$?

$$If; \quad s < 0 \Rightarrow -s > 0 \Rightarrow -s = m\ (say) \in R^+$$

so, then equation (1) becomes

$$\lim_{T\to\infty} \frac{1}{e^{st}} = \lim_{T\to\infty} e^{-st} = \lim_{T\to\infty} e^{mt} \Rightarrow e^{\infty} = \infty$$

. whenever $s < 0$, we cannot calculate the transform of "1". Hence, we can take only $s >$

Laplace Transform of Exponential Function:

Let $f(t) = e^{at}$. Here "a" is a constant, then evaluate $L(e^{at})$

Solution: Laplace transform for given function,

$$L(e^{at}) = \int_0^{\infty} e^{-st}\, e^{at}\, dt \Rightarrow \int_0^{\infty} e^{-(s-a)t}\, dt$$

Integrating and applying limits, we have

$$L(e^{at}) = \left|\frac{e^{-(s-a)t}}{-(s-a)}\right|_0^{\infty} \Rightarrow \frac{-1}{(s-a)}\left[e^{-(s-a)t}\right]_0^{\infty}$$

$$L(e^{at}) = \frac{-1}{(s-a)}\left[\frac{1}{e^{(s-a)t}}\right]_0^{\infty} = \frac{-1}{(s-a)}\left[\frac{1}{e^{\infty}} - \frac{1}{e^{0}}\right]$$

$$L(e^{at}) = \frac{1}{(s-a)}$$

This is our required result. $(For\ s-a>0 \Rightarrow s>a)$

$\therefore L(e^{at}) = \frac{1}{s-a}$ and $L(e^{-at}) = \frac{1}{s+a}$, so

$$L(\cosh at) = \frac{1}{2}\left[\frac{s+a+s-a}{s^2-a^2}\right] \Rightarrow \frac{s}{s^2-a^2}$$

This is our required result.

LECTURE NO. 80

Laplace Transform of Cosine and Sine:

Show that

$$L(\cos at) = \frac{s}{s^2+a^2} \quad \text{and} \quad L(\sin at) = \frac{a}{s^2+a^2}$$

Proof: Suppose that

$$L(\cos at) = F_c \quad \text{and} \quad L(\sin at) = F_s$$

$\boldsymbol{L(\cos at)}$: Applying Laplace Transform

$$F_c = \int_0^\infty e^{-st} (\cos at)\, dt$$

$$\frac{s}{a}F_s = \frac{1}{s} - \frac{a}{s}F_s \quad \Rightarrow \quad \left(\frac{s}{a} + \frac{a}{s}\right)\frac{s}{a}F_s = \frac{1}{s}$$

$$\left(\frac{s^2 + a^2}{as}\right)F_s = \frac{1}{s} \quad \Rightarrow \quad F_s = \left(\frac{a}{s^2 + a^2}\right)$$

As $L(sin\ at) = F_s$, so

$$L(sin\ at) = \frac{a}{s^2 + a^2}$$

This is our required result.

LECTURE NO. 81

Laplace Transform of t^n $(n \in N)$:

Theorem: Prove that

$$L(t^n) = \frac{n!}{s^{n+1}} \quad ; \quad n \in N \qquad \ldots (1)$$

Proof: We'll prove it by mathematical induction.

For $n = 0$;

$$L(t^0) = \frac{0!}{\quad} \Rightarrow \quad L(1) = \frac{1}{\quad}$$

$$L(t^{n+1}) = -\frac{1}{s}\left[\frac{t^{n+1}}{e^{st}}\right]_0^{\infty} + \frac{n+1}{s}\int_0^{\infty} e^{-st}\, t^n\, dt$$

$$L(t^{n+1}) = -\frac{1}{s}\left[\lim_{t\to\infty}\frac{t^{n+1}}{e^{st}} - 0\right] + \frac{n+1}{s}\,L(t^n)$$

As given $L(t^n) = \frac{n!}{s^{n+1}}$, so

$$L(t^{n+1}) = -\frac{1}{s}[0-0] + \frac{n+1}{s}\left(\frac{n!}{s^{n+1}}\right)$$

$$L(t^{n+1}) = \frac{(n+1)!}{s^{n+2}}$$

I.e. it is true for "$n+1$". Hence, proved our required result.

LECTURE NO. 82

Laplace Transform of t^{α} $(\alpha > -1)$:

Aaplying Laplace Transorm,

$$L(t^{\alpha}) = \int_0^{\infty} e^{-st}\, t^{\alpha}\, dt$$

LECTURE NO. 82

Laplace Transform of t^{α} $(\alpha > -1)$:

Aaplying Laplace Transorm,

$$L(t^{\alpha}) = \int_0^{\infty} e^{-st}\ t^{\alpha}\ dt$$

Put $\begin{cases} v = st \Rightarrow dv = sdt \Rightarrow dt = \frac{1}{s}dv \text{ and } t = \frac{s}{v} \\ As\ t \to 0 \Rightarrow v \to 0 \text{ and as } t \to \infty \Rightarrow v \to \infty \end{cases}$

$$L(t^{\alpha}) = \int_0^{\infty} e^{-v} \left(\frac{v}{s}\right)^{\alpha} \frac{1}{s}\, dv \Rightarrow \int_0^{\infty} e^{-v} \left(\frac{v^{\alpha}}{s^{\alpha+1}}\right) dv$$

$$L(t^{\alpha}) = \frac{1}{s^{\alpha+1}} \int_0^{\infty} e^{-v}\, v^{\alpha}\, dv$$

Here

$$\int_0^{\infty} e^{-v}\, v^{\alpha}\, dv = \text{Gamma Function} = \gamma(\alpha + 1)$$

So,

$$L(t^{\alpha}) = \frac{\gamma(\alpha + 1)}{s^{\alpha+1}}$$

This is our required result.

LECTURE NO. 83

S-Shifting Theorem:

$$L(f(t)) = \int_0 e^{-x^2} \frac{\quad}{\sqrt{s}} dx \Rightarrow \frac{\quad}{\sqrt{s}} \int_0 e^{-x^2} dx \quad \ldots(1)$$

Here

$$\int_0^\infty e^{-x^2} dx = \frac{\sqrt{\pi}}{2} \quad (Gaussian\ Integral)$$

Hence relation (1) becomes,

$$L(f(t)) = \frac{2}{\sqrt{s}} \times \frac{\sqrt{\pi}}{2} = \sqrt{\frac{\pi}{s}} \; ; \; s > 0$$

Hence $\frac{1}{\sqrt{t}}$ is not a piecewise continuous function but its Laplace Transform exists. This is our required result.

LECTURE NO. 88

Laplace Transform of Derivatives:

Prove that

i) $L(f') = sL(f) - f(0)$ *ii)* $L(f'') = s^2L(f) - sf(0) - f'(0)$

Proof: since we know

$$If;\ f\ is\ differentiable \Rightarrow f\ is\ continuous.$$

I):

$f(t)$	$F(s) = \mathcal{L}[f(t)]$		Formula
$f(t) = 1$	$F(s) = \frac{1}{s}$	$s > 0$	A
$f(t) = e^{at}$	$F(s) = \frac{1}{(s-a)}$	$s > a$	B
$f(t) = t^n$	$F(s) = \frac{n!}{s^{(n+1)}}$	$s > 0$	C
$f(t) = \sin(at)$	$F(s) = \frac{a}{s^2 + a^2}$	$s > 0$	D
$f(t) = \cos(at)$	$F(s) = \frac{s}{s^2 + a^2}$	$s > 0$	E
$f(t) = \sinh(at)$	$F(s) = \frac{a}{s^2 - a^2}$	$s > \lvert a \rvert$	F
$f(t) = \cosh(at)$	$F(s) = \frac{s}{s^2 - a^2}$	$s > \lvert a \rvert$	G
$f(t) = t^n e^{at}$	$F(s) = \frac{n!}{(s-a)^{(n+1)}}$	$s > a$	H
$f(t) = e^{at}\sin(bt)$	$F(s) = \frac{b}{(s-a)^2 + b^2}$	$s > a$	I
$f(t) = e^{at}\cos(bt)$	$F(s) = \frac{(s-a)}{(s-a)^2 + b^2}$	$s > a$	J
$f(t) = e^{at}\sinh(bt)$	$F(s) = \frac{b}{(s-a)^2 - b^2}$	$s - a > \lvert b \rvert$	K
$f(t) = e^{at}\cosh(bt)$	$F(s) = \frac{(s-a)}{(s-a)^2 - b^2}$	$s - a > \lvert b \rvert$	L

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^\infty e^{-st} f(t)\,dt = \lim_{b\to\infty} \int_0^b e^{-st} f(t)\,dt$$

Table of Laplace Transforms

	$f(t)=\mathcal{L}^{-1}\{F(s)\}$	$F(s)=\mathcal{L}\{f(t)\}$		$f(t)=\mathcal{L}^{-1}\{F(s)\}$	$F(s)=\mathcal{L}\{f(t)\}$
1.	1	$\frac{1}{s}$	2.	e^{at}	$\frac{1}{s-a}$
3.	$t^n,\ n=1,2,3,\ldots$	$\frac{n!}{s^{n+1}}$	4.	$t^p, p>-1$	$\frac{\Gamma(p+1)}{s^{p+1}}$
5.	$\sqrt{t}$	$\frac{\sqrt{\pi}}{2s^{\frac{3}{2}}}$	6.	$t^{n-\frac{1}{2}},\ n=1,2,3,\ldots$	$\frac{1\cdot3\cdot5\cdots(2n-1)\sqrt{\pi}}{2^n s^{n+\frac{1}{2}}}$
7.	$\sin(at)$	$\frac{a}{s^2+a^2}$	8.	$\cos(at)$	$\frac{s}{s^2+a^2}$
9.	$t\sin(at)$	$\frac{2as}{(s^2+a^2)^2}$	10.	$t\cos(at)$	$\frac{s^2-a^2}{(s^2+a^2)^2}$
11.	$\sin(at)-at\cos(at)$	$\frac{2a^3}{(s^2+a^2)^2}$	12.	$\sin(at)+at\cos(at)$	$\frac{2as^2}{(s^2+a^2)^2}$
13.	$\cos(at)-at\sin(at)$	$\frac{s(s^2-a^2)}{(s^2+a^2)^2}$	14.	$\cos(at)+at\sin(at)$	$\frac{s(s^2+3a^2)}{(s^2+a^2)^2}$
15.	$\sin(at+b)$	$\frac{s\sin(b)+a\cos(b)}{s^2+a^2}$	16.	$\cos(at+b)$	$\frac{s\cos(b)-a\sin(b)}{s^2+a^2}$
17.	$\sinh(at)$	$\frac{a}{s^2-a^2}$	18.	$\cosh(at)$	$\frac{s}{s^2-a^2}$
19.	$e^{at}\sin(bt)$	$\frac{b}{(s-a)^2+b^2}$	20.	$e^{at}\cos(bt)$	$\frac{s-a}{(s-a)^2+b^2}$
21.	$e^{at}\sinh(bt)$	$\frac{b}{(s-a)^2-b^2}$	22.	$e^{at}\cosh(bt)$	$\frac{s-a}{(s-a)^2-b^2}$
23.	$t^n e^{at},\ n=1,2,3,\ldots$	$\frac{n!}{(s-a)^{n+1}}$	24.	$f(ct)$	$\frac{1}{c}F\left(\frac{s}{c}\right)$
25.	$u_c(t)=u(t-c)$ Heaviside Function	$\frac{e^{-cs}}{s}$	26.	$\delta(t-c)$ Dirac Delta Function	e^{-cs}
27.	$u_c(t)f(t-c)$	$e^{-cs}F(s)$	28.	$u_c(t)g(t)$	$e^{-cs}\mathcal{L}\{g(t+c)\}$
29.	$e^{ct}f(t)$	$F(s-c)$	30.	$t^n f(t),\ n=1,2,3,\ldots$	$(-1)^n F^{(n)}(s)$
31.	$\frac{1}{t}f(t)$	$\int_s^\infty F(u)\,du$	32.	$\int_0^t f(v)\,dv$	$\frac{F(s)}{s}$
33.	$\int_0^t f(t-\tau)g(\tau)\,d\tau$	$F(s)G(s)$	34.	$f(t+T)=f(t)$	$\frac{\int_0^T e^{-st}f(t)\,dt}{1-e^{-sT}}$
35.	$f'(t)$	$sF(s)-f(0)$	36.	$f''(t)$	$s^2F(s)-sf(0)-f'(0)$
37.	$f^{(n)}(t)$	$s^nF(s)-s^{n-1}f(0)-s^{n-2}f'(0)\cdots-sf^{(n-2)}(0)-f^{(n-1)}(0)$			

Table of Laplace Transforms

	$f(t)=\mathcal{L}^{-1}\{F(s)\}$	$F(s)=\mathcal{L}\{f(t)\}$		$f(t)=\mathcal{L}^{-1}\{F(s)\}$	$F(s)=\mathcal{L}\{f(t)\}$
1.	1	$\frac{1}{s}$	2.	e^{at}	$\frac{1}{s-a}$
3.	$t^n,\ n=1,2,3,\ldots$	$\frac{n!}{s^{n+1}}$	4.	$t^p, p>-1$	$\frac{\Gamma(p+1)}{s^{p+1}}$
5.	$\sqrt{t}$	$\frac{\sqrt{\pi}}{2s^{\frac{3}{2}}}$	6.	$t^{n-\frac{1}{2}},\ n=1,2,3,\ldots$	$\frac{1\cdot3\cdot5\cdots(2n-1)\sqrt{\pi}}{2^n s^{n+\frac{1}{2}}}$
7.	$\sin(at)$	$\frac{a}{s^2+a^2}$	8.	$\cos(at)$	$\frac{s}{s^2+a^2}$
9.	$t\sin(at)$	$\frac{2as}{(s^2+a^2)^2}$	10.	$t\cos(at)$	$\frac{s^2-a^2}{(s^2+a^2)^2}$
11.	$\sin(at)-at\cos(at)$	$\frac{2a^3}{(s^2+a^2)^2}$	12.	$\sin(at)+at\cos(at)$	$\frac{2as^2}{(s^2+a^2)^2}$
13.	$\cos(at)-at\sin(at)$	$\frac{s(s^2-a^2)}{(s^2+a^2)^2}$	14.	$\cos(at)+at\sin(at)$	$\frac{s(s^2+3a^2)}{(s^2+a^2)^2}$
15.	$\sin(at+b)$	$\frac{s\sin(b)+a\cos(b)}{s^2+a^2}$	16.	$\cos(at+b)$	$\frac{s\cos(b)-a\sin(b)}{s^2+a^2}$
17.	$\sinh(at)$	$\frac{a}{s^2-a^2}$	18.	$\cosh(at)$	$\frac{s}{s^2-a^2}$
19.	$e^{at}\sin(bt)$	$\frac{b}{(s-a)^2+b^2}$	20.	$e^{at}\cos(bt)$	$\frac{s-a}{(s-a)^2+b^2}$
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23.	$t^n e^{at},\ n=1,2,3,\ldots$	$\frac{n!}{(s-a)^{n+1}}$	24.	$f(ct)$	$\frac{1}{c}F\left(\frac{s}{c}\right)$
25.	$u_c(t)=u(t-c)$ Heaviside Function	$\frac{e^{-cs}}{s}$	26.	$\delta(t-c)$ Dirac Delta Function	e^{-cs}
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29.	$e^{ct}f(t)$	$F(s-c)$	30.	$t^n f(t),\ n=1,2,3,\ldots$	$(-1)^n F^{(n)}(s)$
31.	$\frac{1}{t}f(t)$	$\int_s^\infty F(u)\,du$	32.	$\int_0^t f(v)\,dv$	$\frac{F(s)}{s}$
33.	$\int_0^t f(t-\tau)g(\tau)\,d\tau$	$F(s)G(s)$	34.	$f(t+T)=f(t)$	$\frac{\int_0^T e^{-st}f(t)\,dt}{1-e^{-sT}}$
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37.	$f^{(n)}(t)$	$s^nF(s)-s^{n-1}f(0)-s^{n-2}f'(0)\cdots-sf^{(n-2)}(0)-f^{(n-1)}(0)$			

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Jis answer
me doubt
ho hvd
confirm
Kr lein...